# DR. PRITOM JYOTI BORA PHD

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## EXECUTIVE SUMMARY

**DATA SCIENTIST ◇ AI/ML SPECIALIST ◇ DATA ANALYTICS & ENGINEERING**

I have over **seven years** of post-PhD research experience in Data Science, Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, Predictive Modelling, Statistical Analysis and Data Analytics. I have strong experience in developing data-driven models and algorithms for solving complex real-world problems, including real-time signal processing, classification, prediction and multi-variable optimisation. My work involves handling complex datasets, data preprocessing, statistical analysis, model development, testing and validation, with applications across multidisciplinary research and engineering. I have strong programming skills in Python, R, SQL, MATLAB and C/C++, with practical experience in Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch and cloud-based AI/ML platforms. I have also worked on ML-integrated real-time monitoring systems and collaborated with multidisciplinary teams to develop practical solutions. Alongside my research, I have published over 43 journal and conference papers, supervised 30 undergraduate and postgraduate projects, and taught Data Analytics and Machine Learning at university level. I am particularly interested in developing reliable, scalable and data-driven solutions that support real world applications and decision making.

## CAREER TRACK

**1. SENIOR POSTDOCTORAL RESEARCH ASSOCIATE**
Loughborough University, UK — Apr 2025 – Present

**2. RESEARCH & TEACHING FELLOW (SENIOR)**
Singapore University of Technology and Design (SUTD), Singapore — Sep 2022 – Mar 2025

- **Teaching Fellow:** Module taught and developed (Jointly): Data Analytics (introduced data driven materials discovery part; materials properties prediction and optimizations; taught general ML algorithms and programming)
  https://www.sutd.edu.sg/course/40-011-data-and-business-analytics-for-esd-students-only/
  Enrolment: 90 students | Teaching score: 4.5/5 & 4.7/5 (Excellent category score:4.5-5).
- Supervised five student project groups, with five students in each group, on industry-oriented projects with *Collins Aerospace, International SOS, and SingHealth*, focusing on manufacturing/process optimization and commercialization of in-house products.

**3. RESEARCH SCIENTIST**,
National University of Singapore (NUS), Singapore — Jul 2021 – Aug 2022

**4. RESEARCH FELLOW**
Nanyang Technological University (NTU), Singapore — Mar 2020 – Jun 2021

**5. POSTDOCTORAL FELLOW**,
Guandong Technion-Israel Institute of Technology (GTIIT), China — Mar 2019 – Feb 2020

## EDUCATION

**Ph.D,** Indian Institute of Science (IISc), India — Aug 2013 – Dec 2019
INDIA RANK of IISc (last 10 years consistently): #1/1000.
**Thesis Title:** Design and Tailoring of Polymer Composite Ultra-Thin Films for Electromagnetic Interference Shielding

**M.Sc.** in Nanoscience & Technology, Tezpur University, India — 2011 – 2013
FIRST CLASS | CGPA:7.68/10| Rank:2nd out of 10 students

**B.Sc.** in Physics, Jorhat Institute of Science and Technology (JIST), India 2008 – 2011

## AWARDS & ACHIEVEMENTS

- Received Bonus 13000.00 SGD (£7000) for outstanding research work, SUTD, Singapore 2024.
- Received Bonus 5000.00 SGD (~£3000) for outstanding research work, NTU, Singapore, 2021.
- Best Research Work & Presentation Award in 2nd International Conference on Advanced Composite Materials (ICACM), December 2019, Beijing, China.
- Best Paper & Presentation Award in 6th International Conference on Electrical Energy & Networks (ICEEN), March 2018, Nanyang Technological University (NTU), Singapore.
- Recipient of prestigious **UK Global Talent Visa** (duly endorsed by the Royal Academy of Engineering).
- Recipient of several International Travel Grants: International Workshop on Advanced Materials (IWAM 2017, 2019 and 2026) and Department of Science and Technology-Science and Engineering Research Board (DST-SERB) International Travel Grant 2017.

## SKILLS

- **Programming: Python**, MATLAB, R, SQL, C/C++, Java.
- **AI/ML**: Artificial Neural Networks (ANN), Deep Learning, Bayesian Networks, Gaussian Process Regression (GPR), Support Vector Machine (SVM), Multi-Objective Bayesian Optimisation (MOBO), Adaptive Neuro-Fuzzy Inference System (ANFIS), Physics-Guided Neural Networks (PGNN), Physics-Informed Neural Networks (PINN).
- **Data Science & Analytics:** Data Preprocessing, Data Cleaning, Feature Engineering, Statistical Analysis, Predictive Modelling, Regression, Classification, Data Visualisation, Multivariable Optimisation and Model Validation.
- **Data Engineering & Management:** Python-based Data Processing, Data Transformation, Dataset Preparation, Data Quality Assessment, SQL, Database Management Fundamentals and Reproducible Data Analysis.
- **AI/ML Frameworks & Libraries:** TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, SciPy and Jupyter Notebook.
- **Cloud Computing & AI Platforms**: AWS SageMaker, Google Vertex AI, Microsoft Azure ML, GPU-Accelerated Machine Learning and Cloud-Based AI Development.
- **Statistical Modelling & Optimisation:** Bayesian Statistics, Probabilistic Inference, Bayesian Optimisation, Gaussian Processes, Uncertainty Analysis and Data-Driven Optimisation.
- **Data Visualisation & Business Intelligence:** Power BI, Tableau, Python (Matplotlib), Statistical Data Interpretation and Reporting.
- **Real-Time Data Analytics:** Signal Processing, Signal Classification, ML-Based Monitoring, Software-Hardware Integration and Real-Time Data-Driven Applications.
- Multiphysics Simulation: ANSYS, ABAQUS, COMSOL and CST Studio Suite.

## PUBLICATIONS

- 1 Book Chapter
- > 10 Invited Talks
- Number of Journal Papers Published: 33
  Out of these, I am the lead/first author on 26 high-impact journal papers**.**
- Submitted journal papers: 3 (I am the lead and corresponding author of all submitted papers)
- Number of Conference Papers Published (IEEE and AIP): 9
- Total Number of Publications (Journal +Conference): > 45
- h-index: 18 Citation count: >1000

Link for all the published papers: https://scholar.google.co.in/citations?user=vCgaxQ4AAAAJ&hl=en&oi=ao

## REFEREES

Prof. Nagarajan Raghavan, SUTD, Singapore. E-mail: nagarajan@sutd.edu.sg
(* Well known for AI/ML including Bayesian and PINN)
Prof. Praveen C. Ramamurthy, IISc, India. E-mail: praveen@iisc.ac.in
Prof. Will Whittow, Loughborough University, UK. E-mail : w.g.whittow@lboro.ac.uk